

SimplePOS SaaS

Detailed Project Report (DPR) — V3

Revision focus: configurable invoice prefix/footer, sales discounts, tax handling, partial payments, mutual-agreement credit, financier/EMI plans, and installment tracking.

1. Executive Summary

SimplePOS is a multi-tenant cloud POS SaaS built around one promise: the fastest and simplest way for small businesses to create invoices, record sales and monitor business performance. Version 1 intentionally skips inventory and product setup by allowing manual item entry.

2. Target Customers

- Sweet Shops
- Grocery Stores
- Bakeries
- Clothing Shops
- Mobile Shops
- Hardware Stores
- Restaurants
- Pharmacies
- Small Retailers

3. Unique Selling Proposition

- Manual sales entry
- Three-click workflow
- Browser Print / Save as PDF
- Invoice regeneration
- Customizable invoice templates
- Clean analytics
- Cloud SaaS
- Flexible payment collection including partial payments and credit/EMI tracking

4. Technology Stack

- Laravel / PHP
- Blade + Tailwind CSS CDN
- Alpine.js / Vanilla JavaScript
- MySQL
- Chart.js / Lucide Icons

5. Architecture

- Multi-tenant using shop_id

- Prepared for employee accounts
- Secure authentication
- Modular Laravel MVC architecture
- Business settings stored per shop
- Payment records separated from sales so partial payments and future collections can be tracked independently

6. Version 1 Modules

- Authentication
- Business Settings
- Dashboard
- Manual Sales
- Draft Sales
- Invoice Generator
- Custom Cash Memo
- Sales History
- Invoice Reprint
- Reports
- Super Admin
- Subscription Management

7. Sales, Discount & Tax Model

Sales must support discounts as either a percentage or a fixed monetary amount. The system should preserve both the entered discount value/type and the final calculated discount amount so invoices, reports and future audits remain reliable.

Recommended sales fields: discount_type = percentage|amount; discount_value = entered percentage or fixed amount; discount = final monetary discount amount; tax_rate = applicable tax percentage (optional); tax = final monetary tax amount; grand_total = final amount payable before considering outstanding balance.

Calculation flow: subtotal → discount → taxable/subtotal-after-discount → tax → grand total. The exact tax basis can be configured by the application's business rules.

8. Payment & Outstanding Balance Model

A sale should not store only a single due amount. Instead, each payment is recorded separately in a sale_payments table. The outstanding balance is calculated from the sale total minus the sum of valid payments. This supports full payment, partial payment, later collection, and multiple payment methods.

- Paid in full: grand total = total payments; outstanding = 0.
- Partially paid: total payments < grand total; outstanding remains collectible.
- Credit / mutual agreement: customer owes the shop directly under an agreed arrangement.
- Financed / EMI: a financier is associated with the transaction and the repayment schedule is tracked separately.
- Payment method remains separate from payment plan. Cash/UPI/card/bank describes how money was received; mutual/financer describes how the outstanding obligation is structured.

9. Payment Plan & EMI Model

Payment plans are optional and attached to a sale. A plan can represent a mutual agreement or a financier arrangement. Installments are generated from the plan and each installment can record its paid amount and status.

- Plan type: mutual | financier
- Financer name: nullable for mutual agreements
- Down payment
- Principal / financed amount
- Total payable under the plan
- Installment amount
- Installment count
- Frequency: weekly | monthly | custom as required
- Start date, status and notes
- Individual installment due dates, paid amounts, status and paid date

10. Invoice Features

- Professional invoice numbers
- Configurable invoice prefix from Business Settings
- Walk-in customer support
- Cash / UPI / Card / Bank / Mixed payments
- Completed / Cancelled / Refunded status
- Partial-payment and outstanding-balance display
- Discount as percentage or fixed amount
- Tax amount and optional tax rate
- Classic, Modern, Thermal & A4 templates
- Browser print & PDF
- Configurable invoice footer text
- Invoice regeneration / reprint

11. Business Settings

- Business Name
- Logo
- Address
- Phone
- FSSAI/GST
- Currency
- Timezone
- Invoice Prefix
- Invoice Template
- Paper Size
- Footer
- Theme Color

12. Analytics & Reports

- Today's Sales
- Invoice Count
- Average Invoice
- Highest Invoice
- Monthly Sales
- Payment Breakdown
- Outstanding / Due Amount
- Recent Sales
- Sales Trend

- Discount Summary
- Tax Summary
- Collection history and installment status

13. Database Tables

- shops
- users
- settings
- customers
- sales
- sale_items
- sale_payments
- payment_plans
- payment_installments
- invoice_sequences
- subscriptions
- activity_logs

14. Updated Database Schema

shops
• id bigint PK
• uuid char(36) unique
• name varchar(150)
• slug varchar(150) unique
• owner_name varchar(100)
• email varchar(150) unique
• phone varchar(30)
• logo varchar(255)
• address text
• city varchar(100)
• state varchar(100)
• country varchar(100)
• status enum(active,inactive,suspended)
• created_at
• updated_at
• deleted_at
users
• id bigint PK
• uuid char(36) unique
• shop_id bigint
• name varchar(100)

• email varchar(150) unique
• phone varchar(30)
• password varchar(255)
• role enum(owner,manager,cashier)
• status enum(active,inactive)
• email_verified_at
• remember_token
• last_login_at
• created_at
• updated_at
• deleted_at
settings
• id bigint PK
• shop_id bigint unique
• business_name varchar(150)
• logo varchar(255)
• phone varchar(30)
• email varchar(150)
• address text
• invoice_prefix varchar(20)
• invoice_template enum(classic,modern,thermal58,thermal80,a4)
• paper_size enum(thermal58,thermal80,a4)
• currency varchar(10)
• timezone varchar(100)
• date_format varchar(30)
• gst varchar(100)
• fssai varchar(100)
• footer_text text
• primary_color varchar(20)
• created_at
• updated_at
customers
• id bigint PK
• uuid char(36) unique
• shop_id bigint
• name varchar(100)
• phone varchar(30)
• email varchar(150)
• address text
• notes text
• created_at
• updated_at
• deleted_at

sales
• id bigint PK
• uuid char(36) unique
• shop_id bigint
• user_id bigint
• customer_id bigint nullable
• invoice_no varchar(50) unique
• invoice_date datetime
• subtotal decimal(12,2)
• discount_type enum(percentage,amount)
• discount_value decimal(12,2)
• discount decimal(12,2) — final monetary discount amount
• tax_rate decimal(5,2) nullable
• tax decimal(12,2) — final monetary tax amount
• grand_total decimal(12,2)
• payment_method enum(cash,upi,bank,card,mixed)
• status enum(draft,completed,cancelled,refunded)
• notes text
• created_at
• updated_at
• deleted_at

15. Payment Tables

sale_payments
• id bigint PK
• sale_id bigint
• amount decimal(12,2)
• payment_method enum(cash,upi,bank,card,mixed)
• paid_at datetime
• notes text nullable
• created_at timestamp
• updated_at timestamp
payment_plans
• id bigint PK
• sale_id bigint unique
• type enum(mutual,financer)
• financer_name varchar(150) nullable
• down_payment decimal(12,2)
• principal_amount decimal(12,2)
• total_payable decimal(12,2)
• installment_amount decimal(12,2)
• installment_count int
• frequency enum(weekly,monthly,custom)
• start_date date
• status enum(active,completed,cancelled)
• notes text nullable
• created_at timestamp
• updated_at timestamp
payment_installments
• id bigint PK
• payment_plan_id bigint
• due_date date
• amount decimal(12,2)
• paid_amount decimal(12,2) default 0
• status enum(pending,partial,paid,overdue)
• paid_at datetime nullable
• created_at timestamp
• updated_at timestamp
invoice_sequences
• id bigint PK
• shop_id bigint
• year int
• month int
• current_number int

• created_at timestamp
• updated_at timestamp
subscriptions
• id bigint PK
• shop_id bigint
• plan_name varchar(100)
• amount decimal(12,2)
• billing_cycle enum(monthly,quarterly,yearly)
• starts_at date
• expires_at date
• payment_reference varchar(255)
• status enum(active,expired,cancelled)
• created_at timestamp
• updated_at timestamp
activity_logs
• id bigint PK
• shop_id bigint
• user_id bigint
• action varchar(150)
• description text
• ip_address varchar(45)
• user_agent text
• created_at timestamp

16. Relationships

- users.shop_id → shops.id
- settings.shop_id → shops.id
- customers.shop_id → shops.id
- sales.shop_id → shops.id
- sales.user_id → users.id
- sales.customer_id → customers.id
- sale_items.sale_id → sales.id
- sale_payments.sale_id → sales.id
- payment_plans.sale_id → sales.id
- payment_installments.payment_plan_id → payment_plans.id
- invoice_sequences.shop_id → shops.id
- subscriptions.shop_id → shops.id
- activity_logs.shop_id → shops.id
- activity_logs.user_id → users.id

17. Example Payment Scenarios

Scenario	Grand Total	Payments	Outstanding
Fully Paid	■50,000	■50,000	■0
Partial Payment	■50,000	■20,000	■30,000
Mutual Agreement	■50,000	■20,000	■30,000 under agreement
Financer / EMI	■50,000	■10,000 down payment	■40,000 financed

18. Version 2 Roadmap

- Products
- Inventory
- Barcode
- Suppliers
- Expenses
- Customer History
- WhatsApp Invoice
- QR Payments
- Multi Branch
- Mobile App

19. Development Roadmap

- Database
- MVC Structure
- Authentication
- Business Settings
- Dashboard
- Sales Module
- Invoice Engine
- Payment / Due Management
- Sales History
- Reports
- EMI / Installment Management
- Super Admin

Schema design note: The existing sales discount and tax monetary fields are retained for final calculated amounts. Discount type/value are added so the original user input is preserved. Payments are normalized into separate records rather than storing only a mutable due amount.